The Spence Loss: Well-Conditioned Value-Function Learning for Reward-Guided Diffusion

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Abstract

Reward-guided diffusion and diffusion fine-tuning are, through entropy-regularized stochastic control, the problem of learning a value function $V(x,t) = \log \mathbb{E}_{\text{base}}[\exp r(X_1) \mid X_t = x]$—a conditional log-partition function whose gradient is the optimal control. The unbiased Monte-Carlo target for V lives in exponential space, so squared-error regression has gradients that explode for over-predictions and, worse, vanish for under-predictions, where the model is most wrong and gradient clipping cannot help. We observe that every Bregman divergence has the value function as its population minimizer, and introduce the *Spence loss*[1]: the Bregman divergence whose gradient has quadratic tails on both sides, with a closed-form numerically-stable gradient. We place it in a family with squared error and the Itakura–Saito divergence. A secondary contribution analyzes why high-dimensional value learning is signal-sparse and studies on-policy importance-sampling schemes. We validate on 2-moons, a dimensional-scaling benchmark designed for clean statistical reads, and GMM-40, where our log-partition estimate is competitive with published diffusion samplers.

1 Introduction

2 value functions for reward-guided diffusion

3 The problem with exponential-space squared error

The value $V(x,t) = \log \mathbb{E}_{\text{base}}[e^{r(X_1)} \mid X_t = x]$ is a conditional log-partition function, and the H-martingale property (§2) supplies *conditionally unbiased* targets in *exponential* space: $e^{V(x_t)} = \mathbb{E}[e^q \mid x_t]$ for a one-step target q (the terminal reward, or a bootstrapped value). A network predicts the value in log space, $p = V_\theta(x_t)$; matching e^p to the unbiased target e^q with squared error gives

$$L_{\text{MSE}}(p,q) = \left(e^p - e^q\right)^2, \qquad \frac{\partial L_{\text{MSE}}}{\partial p} = 2e^p\left(e^p - e^q\right) = 2e^{2p} - 2e^{p+q}. \tag{1}$$

Both tails are pathological, but asymmetrically so (Fig. 1, left).

Over-prediction explodes (recoverable). As $p \to +\infty$ the gradient grows like $2e^{2p}$ and overflows single precision once $p \gtrsim 44$ nats (the square needs e^{2p}). This is a nuisance but a benign one: the gradient merely needs to be a full-size, correctly-signed step, so saturating it to a large finite value (or clipping) recovers a usable update.

Under-prediction vanishes (unrecoverable). As $p \to -\infty$ the gradient behaves like $-2e^{p+q} \to 0$. This is exactly the regime where the model is *most wrong*: it assigns near-zero value $e^p \approx 0$ to a state whose true value e^q is large — a rare, high-reward target it has failed to find. The learning signal there is not merely ill-scaled, it is *absent*, and no amount of gradient clipping can manufacture a gradient that has decayed to zero. In the high-dimensional, signal-sparse regime of §6, where the useful targets are precisely the rare large ones, this is the dominant failure mode.

[1] Named for the Spence (dilogarithm) function Li_2 appearing in its value; in our code and experiments it is the `quad` loss.

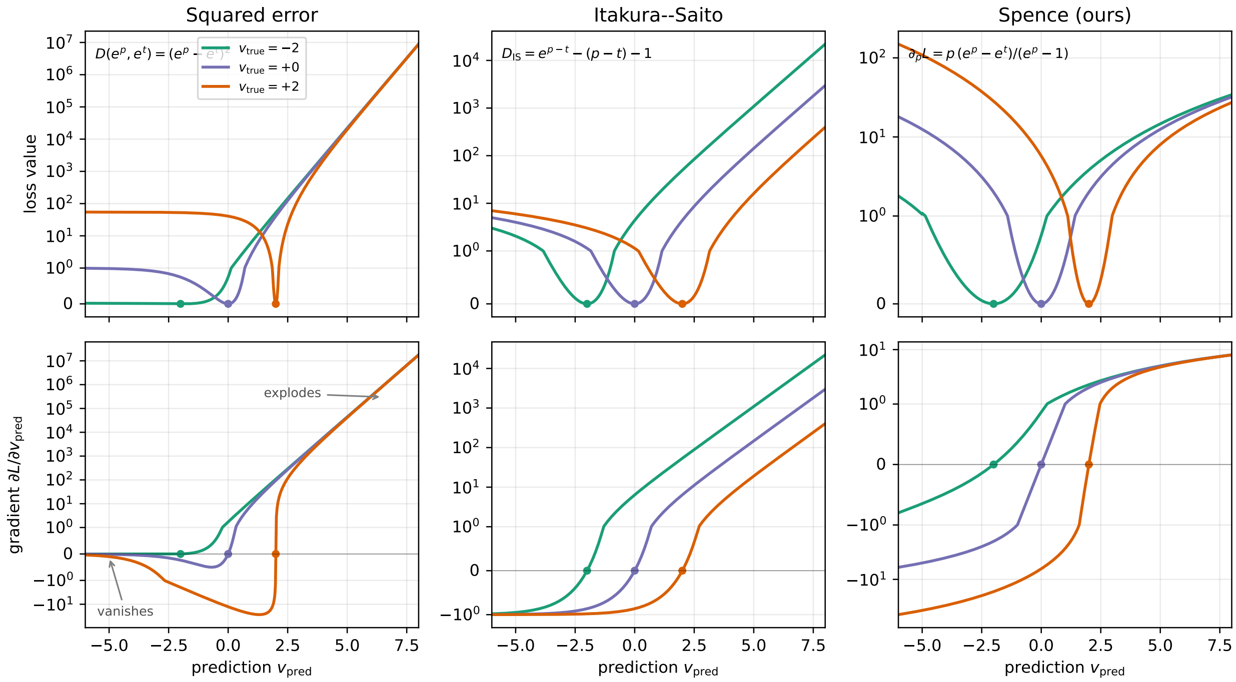

Figure 1: Loss value (top) and gradient $\partial L/\partial v_{\text{pred}}$ (bottom) of $D_F(e^{v_{\text{pred}}}, e^{v_{\text{true}}})$ as functions of the prediction v_{pred}, for three targets $v_{\text{true}} \in \{-2, 0, 2\}$ (dots mark the minimizer $v_{\text{pred}} = v_{\text{true}}$; symmetric-log axes). **Squared error** vanishes for under-predictions ($v_{\text{pred}} < v_{\text{true}}$, gradient $\to 0$) and explodes for over-predictions. **Itakura–Saito** bounds the under-prediction gradient at -1 (removing the vanishing pathology) but still explodes for over-predictions. **Spence** (ours) is quadratic on both sides: the gradient is linear in v_{pred} everywhere, so neither tail vanishes nor explodes exponentially.

Empirical pathology. Beyond the asymptotics, the finite-precision behavior is severe. In our convergence runs the naive loss skipped or aborted a large fraction of batches on overflow (e.g. 18% of batches in one FBRRT run once predictions reached moderate magnitude), and a *single* non-finite prediction propagated through the shared value network to poison every subsequent batch. A stabilized implementation with the exact gradient of (1) saturated to the float range (our `exp_mse`) removes the overflow and the poisoning, but it cannot remove the vanishing gradient, which is a property of the divergence, not the arithmetic. Fixing it requires changing the loss while keeping the unbiased exponential-space target — exactly what the Bregman family affords (§4).

4 Bregman divergences for value learning

4.1 Any Bregman divergence learns the value function

For a strictly convex, differentiable potential φ, the Bregman divergence is

$$D_\varphi(u, v) = \varphi(u) - \varphi(v) - \varphi'(v)\,(u - v). \tag{2}$$

Squared error is the special case $\varphi(u) = u^2$. The property we exploit:

Lemma 1 (Bregman divergences are proper losses for the value)**.** *Let the regression target be e^q with q a conditionally-unbiased estimate of the value (the H-martingale target, §2). For any strictly convex φ,*

$$\arg\min_u \; \mathbb{E}_{q|x_t}\big[D_\varphi(u,\, e^q)\big] \;=\; \mathbb{E}[e^q \mid x_t], \tag{3}$$

so the minimizing prediction in log space is $p^\star = \log \mathbb{E}[e^q \mid x_t] = V(x_t)$.

Thus the choice of φ does *not* bias the learned value; it is free to be chosen for *conditioning*. (Proof in App. D.)

4.2 Design criteria and two reference points

We parameterize the prediction in log space, $u = e^p$, and ask how the gradient $\partial L/\partial p$ behaves as the log-error $p - q \to \pm\infty$.

Squared error ($\varphi(u) = u^2$). $\partial L/\partial p = 2e^p(e^p - e^q) = 2e^{2p} - 2e^{p+q}$, which *explodes* as $p \to +\infty$ and *vanishes* as $p \to -\infty$ — the under-prediction pathology of §3.

Itakura–Saito ($\varphi(u) = -\ln u$). With the prediction as the first argument (Lemma 1), $D_{\mathrm{IS}}(e^p, e^q) = e^{p-q} - (p - q) - 1$, whose gradient in p is

$$\frac{\partial L_{\mathrm{IS}}}{\partial p} = e^{p-q} - 1. \tag{4}$$

As $p \to -\infty$ this tends to -1: a bounded, *non-vanishing* corrective signal for under-predictions, exactly removing squared error's worst pathology. It still explodes as $p \to +\infty$ (e^{p-q}), but that is the recoverable, clippable direction. Itakura–Saito is moreover *scale-invariant* ($D_{\mathrm{IS}}(cu, cv) = D_{\mathrm{IS}}(u, v)$) and is the negative log-likelihood of a gamma/exponential observation model, giving a natural uncertainty interpretation for V. A useful midpoint — it fixes the harder tail — but one-sided (Fig. 1, middle).

4.3 The Spence loss

We propose the Bregman divergence generated by the potential F with

$$F''(u) = -\frac{\ln u}{u\,(1 - u)}, \tag{5}$$

applied to $u = e^p$ against the target e^q. Its gradient with respect to the prediction has the closed form

$$\boxed{\frac{\partial L}{\partial p} \;=\; p\,\frac{e^p - e^q}{e^p - 1}} \tag{6}$$

which is zero iff $p = q$ and, by Lemma 1, has population minimizer V.

Two-sided quadratic tails. From (6),

$$p \to +\infty: \quad \frac{\partial L}{\partial p} \to p \qquad \text{(no } e^{2p} \text{ blow-up; over-prediction is quadratically penalized),} \tag{7}$$

$$p \to -\infty: \quad \frac{\partial L}{\partial p} \to p\,e^q \qquad \text{(non-vanishing corrective signal for under-prediction).} \tag{8}$$

Both tails are linear in p, i.e. the loss is quadratic on both sides — the first loss to fix both pathologies of exponential-space regression at once.

Value via the Spence function. The divergence value (used only for monitoring; the gradient uses (6)) is expressible through the Spence function $S(x) = \mathrm{Li}_2(1 - e^x)$, using the dilogarithm identity $S'(x) = -\,xe^x/(e^x - 1) = x/\mathrm{expm1}(-x)$; hence the name. Full derivation, the numerically stable branch-wise gradient (finite for all float inputs, unlike the naive autograd which NaNs for $p \gtrsim 88$), and the gamma-like uncertainty interpretation are in App. A.

Table 1: The value-learning loss family. Tails give $\partial L/\partial p$ as the log-error $\to \pm\infty$. All three share the same minimizer, $V = \log \mathbb{E}[e^q \mid x_t]$ (Lemma 1).

loss	under-predict ($p \to -\infty$)	over-predict ($p \to +\infty$)	scale-inv.	uncertainty model
squared error	vanishes ($\to 0$)	explodes (e^{2p})	no	Gaussian (exp)
Itakura–Saito	bounded ($\to -1$)	explodes (e^{p-q})	yes	gamma
Spence (ours)	**quadratic** ($p\, e^q$)	**quadratic** (p)	no	gamma-like

5 the loss in isolation

6 Signal sparsity and on-policy learning

7 Experiments

8 Related work

9 Conclusion and limitations

References

A The Spence loss: derivation and implementation

B On-policy method zoo

C Experimental details

D Additional proofs